System Engineering T&B
Aachen, Nordrhein-Westfalen
Deutschland

Accessory Belt Drive System
Summary of Results

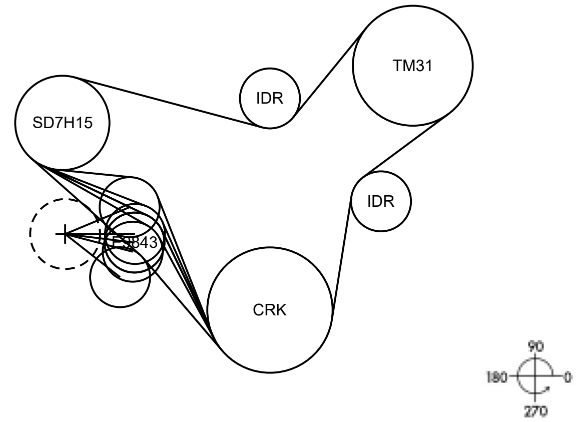
AG00894 BMC 6 Cyln.
TM1 and SD7H15 AC Drive
E9843 (23Nm): CCW@-263.8/97.9
8PK1738HD Poly. Fleetrunner
18Sep2023

Belt Data, DB Ver. = 2.18.0.0, Belt Rev. = 7/14/2016

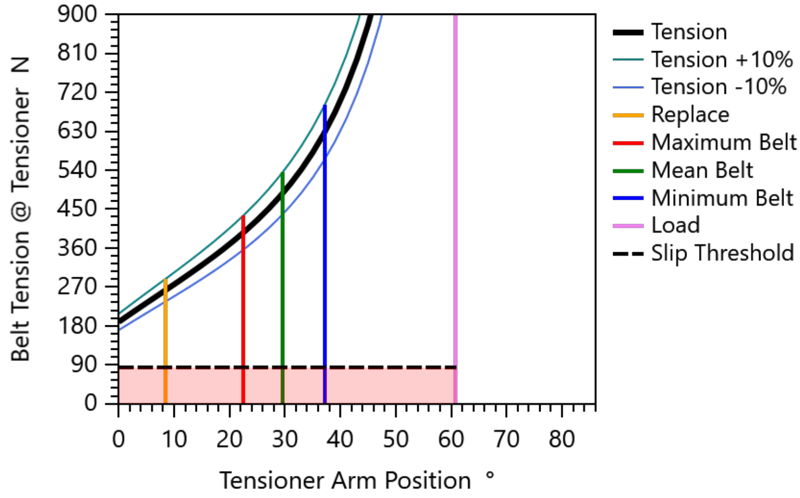
Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Effective Belt Length mm (ISO 9981)	1739

Tensioner Data

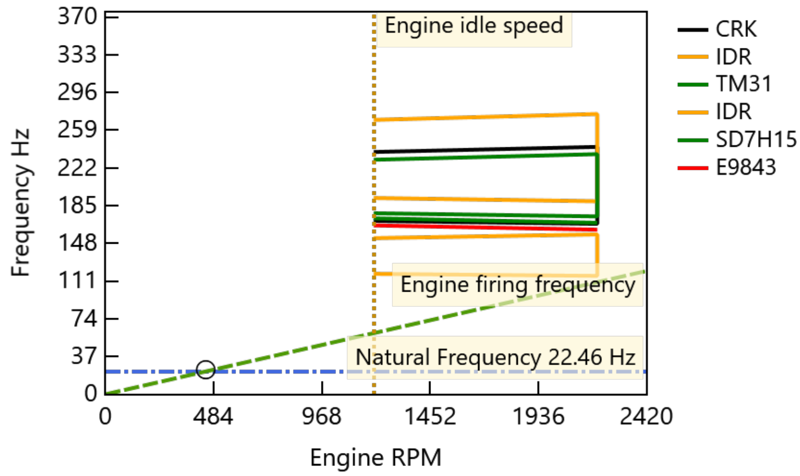
Type	Automatic
Design Tension N	487 (109 lbf)



Belt Tension Control



Natural Frequency Map



Belt Life B10

Duty Cycle DCn-th %	Av. Vehicle Speed km/hr	PK-2_2p-MT3 hours
95	1	8294

Peak Tension & Hubload

	Engine RPM	Accel. RPM/s	Tension N	Hubload N	Direction /Wrap °
CRK	1731	1000	1167	1591.5	91 / 145
IDR	1731	1000	1166	881.9	329 / 44
TM31	1731	1000	982	2131.8	223 / 194
IDR	1731	1000	981	1064.1	108 / 66
SD7H15	1731	1000	488	1463.6	342 / 170
E9843	1200	1000	487	324.1	225 / 39

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)

Permissible Grooved Pulley Axial Offset Allowance
For 0.90° Belt Entry Angle

Adjacent Grooved Pulleys	Flat Pulley Angular Misalignment °	Axial Offset Allowance mm	Grooved Pulley Axial Offset %	Attributed to Flat Pulley Angular Misalignment %
TM31->CRK	IDR (0.33)	2.65	71	29
SD7H15->TM31	IDR (0.33)	2.25	50	50
CRK->SD7H15	E9843 (1.00)	0.49	2	98

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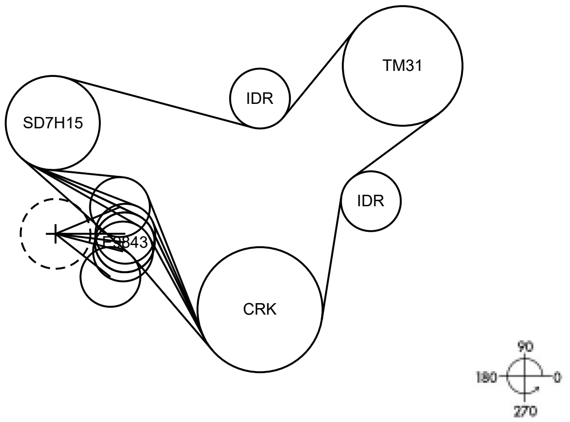
Layout Data mm

	X	Y	Flat	Pitch	Effective	DOB
CRK	0.00	0.00		161.40	159.00	159.99
IDR	143.00	140.00	75.00	77.20	79.60	
TM31	184.00	314.60		154.40	152.00	152.99
IDR	0.00	272.50	75.00	77.20	79.60	
SD7H15	-267.40	241.00		121.40	119.00	119.81
E9843	-174.47	86.93	75.00	77.20	79.60	



Belt Data, DB Ver. = 2.18.0.0, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Stretch and Wear Allowance % of Length	0.70
Belt Length Tolerance (+/-) mm	6.00

Tensioner Data

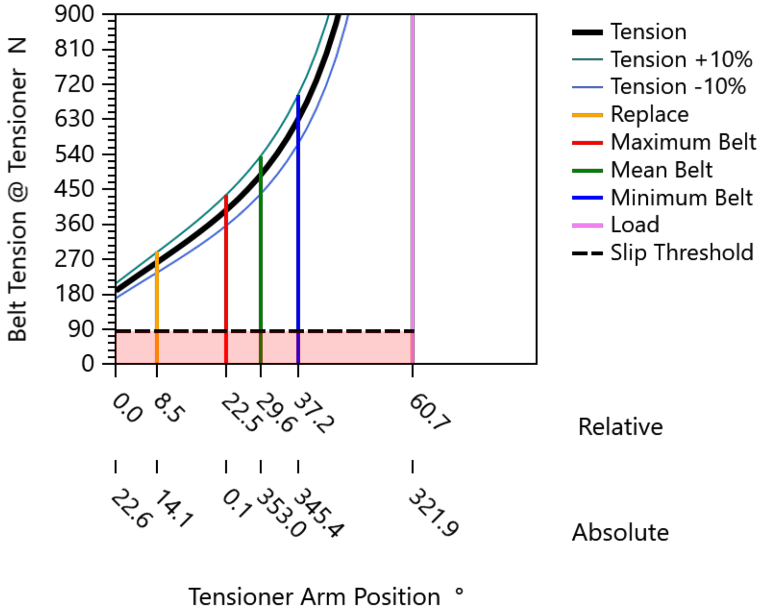
Type	Automatic
Design Tension N	487 (109 lbf)
Pivot Point {X, Y Coordinates} mm	-263.80 , 97.90
Arm Length mm	90.0
Spring Mean Load Nm	23.00
Spring Pre-Load Nm	8.93
Spring Rate Nm/deg	0.475

RESULTS

Belt Drive System Geometry

	Span Length mm	Wrap Angle °	Speed Ratio (Ref. Engine)
CRK	160.7	145.2	1.000
IDR	137.0	44.4	2.091
TM31	149.1	194.2	1.045
IDR	250.3	65.7	2.091
SD7H15	150.0	169.5	1.329
E9843	154.2	38.8	2.091

Belt Tension Control



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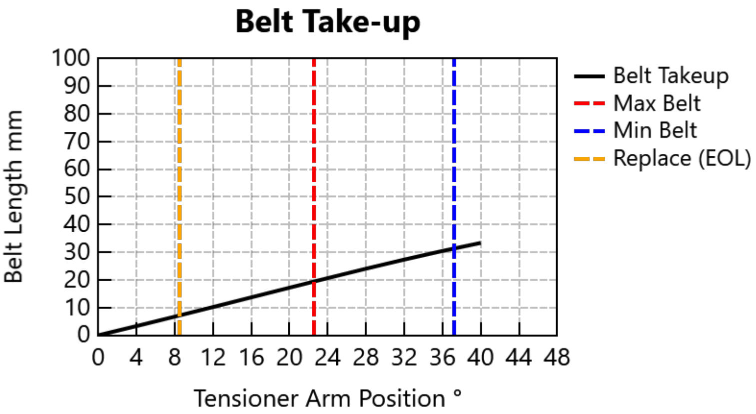
Tensioner Geometry

Belt Takeup / Tensioner Arm Ratio mm/° 0.825

		Belt-Drive Length				
Position Description	**Free Arm**	**Replace**	**Max**	**Mean**	**Min**	**Load**
Arm Position °	22.6	14.1	0.1	353.0	345.4	321.9
Idler X Coordinate mm	-180.7	-176.5	-173.8	-174.5	-176.7	-193.0
Idler Y Coordinate mm	132.5	119.8	98.0	86.9	75.2	42.3
Hubload to Arm Angle °	30.3	35.7	46.2	52.2	59.1	82.6
Hubload Direction °	232.9	229.8	226.3	225.2	224.5	224.5
Hubload N	196.6	246.8	302.2	323.5	344.8	423.5
Tension N	189.0	262.8	395.9	486.7	629.3	3054.8
Idler Wrap Angle °	62.7	56.0	44.9	38.8	31.8	7.9
Effective Drive Length mm	1765.0	1757.9	1745.6	1739.6	1733.6	1721.8
Required Eff. Belt Length mm	1764.1	1756.9	1744.7	1738.7	1732.7	1720.8



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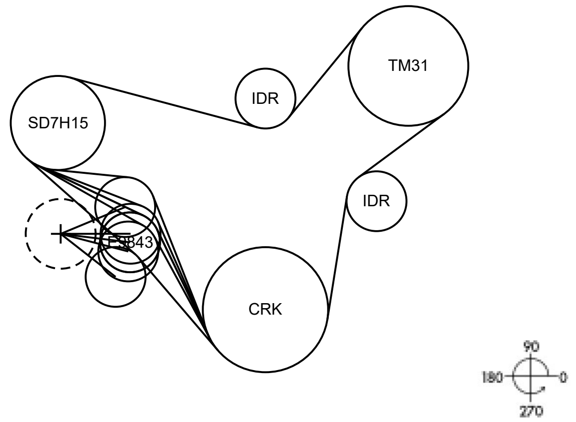
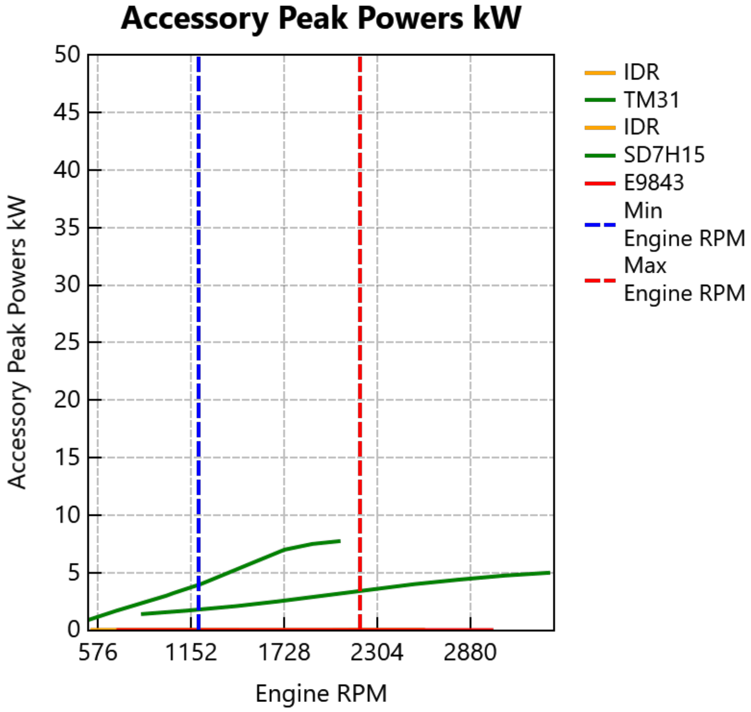
Design Tension N	487 (109 [lbf])
Engine Acceleration RPM/s	1000
Engine Deceleration RPM/s	1000





Accessory Data

	File Name - if used	Moment of Inertia kg m^2
IDR	IDR.cmp	0.0002
TM31	A_C-TM31.cmp	0.0053
IDR	Gpdpidr.IDR	0.0002
SD7H15	A_C-SD7H15.cmp	0.0060
E9843	Gpdpidr.IDR.TEN	0.0002

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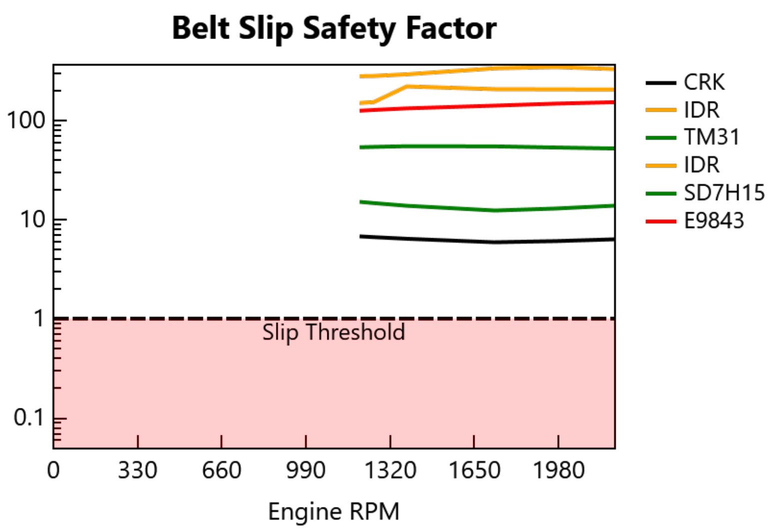
RESULTS

Belt Slip Sensitivity

Pulley	Most Critical Load Condition						Required Increase Tension or Wrap Angle	
	Acceler. RPM/s	Load Combination %						
		TM31	SD7H15				N	°
CRK	1000	100	100					
IDR	1000	10	10					
TM31	1000	100	10					
IDR	1000	100	10					
SD7H15	1000	100	100					
E9843	1000	100	100					



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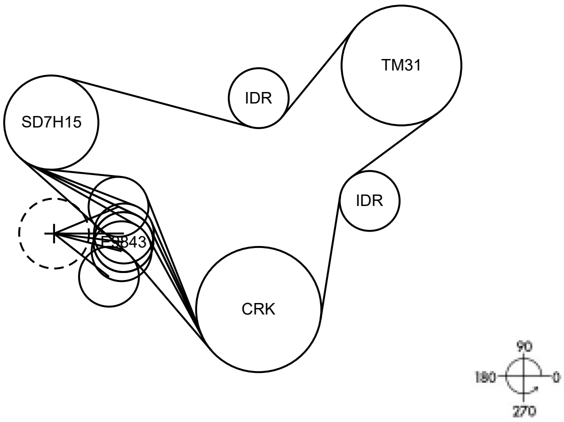

Load Conditions kW for DC 95%

%	Eng. RPM	V, km/hr	T, °C	CRK	IDR	TM31	IDR	SD7H15	E9843
26.60	519	1	90	2.33	0.01	1.40	0.01	0.90	0.01
10.00	693	1	90	3.43	0.01	1.70	0.01	1.70	0.01
13.00	1000	1	90	5.13	0.01	2.10	0.01	3.00	0.01
17.00	1039	1	90	5.73	0.01	2.50	0.01	3.20	0.01
18.00	1212	1	90	7.03	0.01	3.00	0.01	4.00	0.01
8.00	1385	1	90	8.53	0.01	3.50	0.01	5.00	0.01
4.00	1558	1	90	10.03	0.01	4.00	0.01	6.00	0.01
3.00	1731	1	90	11.43	0.01	4.40	0.01	7.00	0.01
0.30	1904	1	90	12.28	0.01	4.75	0.01	7.50	0.01
0.10	2077	1	90	12.78	0.01	5.00	0.01	7.75	0.01

Mean Hubloads N for DC 95%

%	Eng RPM	CRK	IDR	TM31	IDR	SD7H15	E9843
26.60	519	1445	769	1700	754	1178	324
10.00	693	1498	810	1837	845	1262	324
13.00	1000	1519	826	1922	915	1326	324
17.00	1039	1564	861	1977	925	1335	324
18.00	1212	1597	886	2036	953	1361	324
8.00	1385	1639	919	2114	992	1397	324
4.00	1558	1671	944	2175	1023	1425	324
3.00	1731	1691	958	2217	1048	1448	324
0.30	1904	1673	945	2187	1034	1435	324
0.10	2077	1638	918	2128	1007	1411	324

Mean Tensions N for DC 95%

%	Eng RPM	CRK	IDR	TM31	IDR	SD7H15	E9843
26.60	519	1018	1016	696	694	489	487
10.00	693	1072	1071	780	779	488	487
13.00	1000	1094	1093	844	843	488	487
17.00	1039	1139	1138	853	852	488	487
18.00	1212	1173	1172	879	878	488	487
8.00	1385	1216	1215	916	915	488	487
4.00	1558	1249	1248	944	943	487	487
3.00	1731	1268	1267	967	966	487	487
0.30	1904	1250	1249	954	953	487	487
0.10	2077	1215	1214	929	929	487	487